Project: Phishing Awareness Training

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PROJECT OVERVIEW

Introduction

Phishing is one of the most common cybersecurity threats. Attackers use fake emails, messages, phone calls, and websites to trick people into sharing confidential information or performing unsafe actions.

Objective of the Project

The objective of this project is to create awareness about phishing attacks and help people recognize suspicious messages and fake websites.

Learning Objectives

After completing this training, learners should be able to:

Understand what phishing is and how it works.

Recognize common warning signs in suspicious messages.

Identify social engineering techniques used by attackers.

Follow safe practices when browsing websites and opening emails. Understand what to do after clicking a suspicious link.

Target Audience

This training is designed for students, employees, and general internet users who want to improve their cybersecurity awareness.

WHAT IS PHISHING?

Phishing is a cyberattack in which attackers pretend to be a trusted person or organization to trick people into revealing sensitive information, clicking malicious links, or downloading harmful files.

Attackers may pretend to be banks, delivery companies, colleges, social media platforms, or well-known businesses.

What Do Attackers Want?

Phishing attackers may try to:

Steal usernames and passwords.

Obtain banking details and payment information.

Trick people into sharing One-Time Passwords (OTPs).

Install malware on computers or mobile phones.

Gain unauthorized access to personal or company accounts.

How Does Phishing Reach People?

HOW PHISHING WORKS

Phishing attacks usually follow a series of steps. Attackers use deception to make victims trust a message and take an unsafe action.

Step 1: Impersonation

The attacker pretends to be a trusted organization, such as a bank, college, delivery

company, or employer.

Step 2: Sending a Fake Message

The attacker sends an email, SMS, or message that looks genuine. It may contain a fake link or malicious attachment.

Step 3: Creating Pressure

The message may claim that an account will be blocked, a payment has failed, or a prize will expire soon. This encourages the victim to act quickly without checking.

Step 4: Tricking the Victim

The victim may click a link, open an attachment, enter a password, share an OTP, or approve an unexpected sign-in request.

Step 5: Misusing the Information

The attacker may use stolen information to access accounts, commit fraud, steal money, or gain access to an organization's systems.

Example

WARNING SIGNS OF PHISHING

Phishing messages can look genuine, but they often contain warning signs that help us identify them.

1. Urgent or Threatening Language

Messages may say "Your account will be blocked today" or "Act immediately." Attackers use urgency to make people panic.

2. Suspicious Sender Address

The sender's email address may contain misspellings, extra characters, or a domain that does not match the real organization.

3. Unexpected Links

A message may ask you to click a link to verify your account, claim a reward, or make a payment.

4. Unexpected Attachments

Unknown attachments may contain malware or lead to unsafe downloads.

5. Requests for Confidential Information

Be suspicious of unexpected requests for passwords, OTPs, banking details, or other sensitive information.

6. Spelling and Grammar Mistakes

CHECKING LINKS AND WEBSITES SAFELY

Before clicking a link or entering personal information on a website, check whether the website is genuine.

1. Check the Website Domain

Look carefully at the website address. Attackers may use addresses that resemble genuine websites but contain misspellings or extra words. For example, a fake website might use a domain that looks similar to a bank's official domain.

2. Do Not Trust a Link Just Because It Looks Genuine

A link may display the name of a trusted organization but redirect to a different website. Check the actual destination before opening it.

3. Understand HTTPS

HTTPS helps protect information exchanged between your browser and a website. However, the padlock symbol does not prove that a website is genuine or trustworthy. Fake websites can also use HTTPS.

4. Visit Official Websites Directly

If you receive an unexpected message from a bank, college, or online service, avoid using the link in that message. Instead, type the known official address into your browser or use the organization's official app.

5. Avoid Entering Sensitive Information

COMMON TYPES OF PHISHING

Phishing attacks can happen through different communication channels. Understanding the common types helps people recognize and avoid them.

1. Email Phishing

Attackers send fake emails pretending to represent trusted organizations. The email may contain malicious links or attachments.

2. Spear Phishing

Spear phishing targets a specific person or organization. Attackers may use personal or workplace information to make their messages appear genuine.

3. Smishing (SMS Phishing)

Smishing uses text messages to trick victims. For example, a fake delivery message may ask the recipient to pay a fee through a suspicious link.

4. Vishing (Voice Phishing)

Vishing happens through phone calls. Attackers may pretend to be bank employees, technical support staff, or government officials to obtain confidential information.

5. Business Email Compromise (BEC)

In BEC attacks, criminals impersonate executives, employees, or business partners to trick people into transferring money or sharing sensitive information.

SOCIAL ENGINEERING TRICKS

Social engineering is a technique in which attackers manipulate people into revealing confidential information or performing unsafe actions.

1. Urgency

Attackers claim that an account will be blocked or a payment must be made immediately.

2. Fear

They may threaten legal action, account suspension, or financial loss to make victims panic.

3. Authority

An attacker may pretend to be a manager, bank employee, police officer, or technical support representative.

4. Curiosity

Messages may contain surprising news, attractive links, or unusual attachments to encourage people to click.

5. Rewards and Offers

Attackers may promise prizes, discounts, scholarships, or free gifts to obtain personal information.

6. Trust and Familiarity

REAL-WORLD PHISHING EXAMPLES

Example 1: Fake Delivery Message

A person receives an SMS claiming that a parcel is delayed and a small fee must be paid immediately through a link.

Warning signs: Unexpected payment request, urgency, and an unfamiliar website.

Safe action: Check the delivery status using the official delivery company's app or website.

Example 2: Fake College Account Alert

A student receives an email claiming that their college account will be disabled unless they sign in through a provided link. **Warning signs:** Threat of account closure, unexpected login link, and possible impersonation.

Safe action: Visit the college's official portal directly or contact the IT department.

Example 3: Fake Workplace Payment Request

An employee receives an email that appears to come from a manager requesting an urgent money transfer.

Warning signs: Unusual payment request and pressure to act quickly.

Safe action: Verify the request with the manager through a separate, trusted communication channel.

Lesson

PHISHING PREVENTION CHECKLIST

Follow these best practices to protect yourself from phishing attacks.

Before Opening a Message

Check the sender's email address or phone number.

Be careful with unexpected messages and urgent requests.

Do not open suspicious attachments.

Before Clicking a Link

Inspect the website address carefully.

Visit official websites directly instead of using unexpected links. Remember that HTTPS does not guarantee a website is genuine.

Protect Your Accounts

Use strong, unique passwords.

Enable multi-factor authentication (MFA).

Never share passwords or OTPs with unexpected callers or messages. Keep your browser, operating system, and security software updated.

At College or Work

Verify unusual requests for payments or confidential information.

IT

WHAT TO DO IF YOU CLICK A PHISHING LINK

If you accidentally click a phishing link, stay calm and take the following steps.

1. Stop Interacting

Close the suspicious website. Do not enter any more information or download additional files.

2. Change Your Password

If you entered your password, change it immediately using the official website or app. Change it on any other accounts where you reused the same password.

3. Protect Your Accounts

If you shared an OTP or approved an unexpected sign-in, contact the service provider and review your account activity and active sessions.

4. Contact Your Bank

If you shared banking information or made a suspicious payment, contact your bank immediately using its official contact details.

5. Scan Your Device

If you downloaded or opened a suspicious file, run a security scan and seek technical assistance if needed.

6. Report the Incident

INTERACTIVE PHISHING AWARENESS QUIZ

Test your knowledge by selecting the best answer to each question.

Question 1

You receive an unexpected email saying your account will close in 10 minutes unless you click a link. What should you do?

A. Click immediately
B. Verify the message through the official website
C. Reply with your password
D. Forward it to your friends

Question 2

Does the HTTPS padlock guarantee that a website is genuine?
A. Yes, always
B. No, fake websites can also use HTTPS
C. Only on mobile phones
D. Only for banking websites

Question 3

What is smishing?
A. Phishing through SMS messages
QUIZ ANSWER KEY

B- Verify the message through the official website. B Fake websites can also use HTTPS.
.
C— Smishing is phishing through SMS messages.
.
D- Never share an OTP; contact the bank through an official channel.

E— Change your password using the official website and secure your account.
CONCLUSION
Phishing is a serious cybersecurity threat that uses

deception to steal information, compromise accounts, and cause financial loss.

By learning to recognize suspicious messages, checking website addresses, protecting passwords, and verifying unexpected requests, people can reduce their risk of becoming victims.

Everyone has a role in maintaining cybersecurity. Stay alert, think before clicking, and report suspicious activity.

Remember: Pause. Inspect. Verify. Report.

A single careful decision can help protect your personal information and digital identity.

REFERENCES AND PROJECT CREDITS

References

The following trusted resources provide additional information about phishing and online safety.

Cybersecurity and Infrastructure Security Agency (CISA) — Recognize and Report Phishing
https://www.cisa.gov/secure-our-world/recognize-and-report-phishing

Google Safety Center
https://safety.google/

Federal Trade Commission (FTC) - How to Recognize and Avoid Phishing Scams
https://consumer.ftc.gov/articles/how-recognize-and-avoid-phishing-scams

Project Credits

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